

Name: _____

Date: _____

VECTORS,

(ADDING AND SUBTRACTING VECTORS)

LO: I can add and subtract column vectors and represent the results on a grid.

Adding and subtracting vectors

A **vector** has magnitude and direction, written as a column vector. To **add vectors**, add the corresponding components. To subtract, subtract components.

Geometrically: Adding vectors places them end to end (triangle rule). The resultant goes from the start of the first to the end of the last.

Quick examples

●	$(3, 2) + (1, 4) = (4, 6)$	add components
●	$(5, 3) - (2, 1) = (3, 2)$	subtract components
●	$a + b$ means travel along a then along b	triangle rule
●	$(3, 2) + (1, 4) = (4, 8)$	add each component separately
●	$(5, 3) - (2, 1) = (3, 4)$	subtract y from y, not add

Key idea

Add or subtract the x-components and y-components separately.

$$(a, b) + (c, d) = (a+c, b+d)$$

Component-wise addition

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - adding two vectors

Find $(3, 2) + (4, -1)$.

Add x-components: $3 + 4 = 7$

Add y-components: $2 + (-1) = 1$

Result: $(7, 1)$

$(7, 1)$

Add each component separately.

Example 2 - subtracting vectors

Find $(5, 6) - (2, 4)$.

Subtract x-components: $5 - 2 = 3$

Subtract y-components: $6 - 4 = 2$

Result: $(3, 2)$

$(3, 2)$

Subtract each component separately.

Example 3 - adding multiple vectors

Find $(1, 3) + (2, -1) + (-4, 5)$.

x: $1 + 2 + (-4) = -1$

y: $3 + (-1) + 5 = 7$

Result: $(-1, 7)$

$(-1, 7)$

Add all x-components, then all y-components.

Example 4 - finding a vector

If $a + b = (7, 3)$ and $a = (4, 1)$, find b.

$b = (7, 3) - (4, 1)$

x: $7 - 4 = 3$

y: $3 - 1 = 2$, so $b = (3, 2)$

$b = (3, 2)$

Rearrange: $b = \text{resultant} - a$.

YOUR TURN – PRACTICE (deliberate practice)

1. Adding vectors

Find each sum.

- a) $(2, 3) + (4, 1)$
- b) $(5, -2) + (1, 6)$
- c) $(-3, 4) + (7, -2)$
- d) $(0, 5) + (-2, -3)$

2. Subtracting vectors

Find each difference.

- a) $(6, 5) - (2, 3)$
- b) $(3, 1) - (5, 4)$
- c) $(-1, 7) - (-3, 2)$
- d) $(4, -2) - (4, -2)$

3. Adding multiple vectors

Find the resultant vector.

- a) $(1, 2) + (3, 4) + (2, -1)$
- b) $(4, -3) + (-2, 5) + (1, 1)$
- c) $(-1, 6) + (3, -4) + (-2, -2)$
- d) $(5, 0) + (0, 3) + (-5, -3)$

4. Finding unknown vectors

Find the unknown vector.

- a) $a + (3, 2) = (7, 5)$. Find a .
- b) $(5, 1) + b = (2, 6)$. Find b .
- c) $a - (2, 3) = (1, -1)$. Find a .
- d) $(4, 7) - b = (1, 3)$. Find b .

5. Mixed vector operations

Solve each problem.

a) $2(3, 1) + (1, 4)$

b) $(5, 2) - 2(1, 3)$

c) $a = (2, -1)$, $b = (4, 3)$. Find $a + 2b$.

d) $p = (3, 5)$, $q = (-1, 2)$. Find $p - q$.

e) If $a + b + c = (0, 0)$, $a = (3, 2)$, $b = (-1, 4)$. Find c .

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $(3, 2) + (1, 4) = (4, 6)$ ☐

Correct answer: _____

Reason: _____

b) $(3, 2) + (1, 4) = (4, 8)$ ☐

Correct answer: _____

Reason: _____

c) Subtracting a vector reverses its direction ☐

Correct answer: _____

Reason: _____

d) $(5, 3) - (2, 1) = (3, 4)$ ☐

Correct answer: _____

Reason: _____

e) The zero vector $(0, 0)$ is the result when a vector is subtracted from itself ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A boat travels along vectors $(5, 2)$, then $(-3, 4)$, then $(1, -6)$. Find the resultant displacement. How far is the boat from its starting point (to 1 dp)?

Expression: _____

Simplified: _____

b) In quadrilateral ABCD, $AB = (4, 1)$, $BC = (2, 3)$, $CD = (-4, -1)$. Find DA. What type of quadrilateral is ABCD? Justify your answer.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

